

Highlights

VitaTouch: Property-Aware Vision–Tactile–Language Model for Robotic Quality Inspection in Manufacturing

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- VitaSet: 186 objects, 52k images across modalities, and 5.1k QA pairs
- Three-stage training: cross-modal alignment, property learning, and LoRA adaptation
- VitaTouch achieves the best scores on HCT and TVL in the public TVL benchmark
- VitaTouch on VitaSet: 89% hardness, 75% roughness, 55% descriptor recall
- LoRA adaptation reaches 92.0% accuracy on 5-category defect recognition

VitaTouch: Property-Aware Vision–Tactile–Language Model for Robotic Quality Inspection in Manufacturing

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Abstract

Quality inspection in smart manufacturing requires recognizing intrinsic material and surface properties beyond visible geometry, yet vision-only approaches struggle to infer these properties under imaging nuisances. To address these limitations, we propose **VitaTouch**, a **property-aware vision–tactile–language (VTL) model** that integrates visual and tactile sensing with language prompts in a unified semantic space, enabling property inference and natural-language attribute descriptions. VitaTouch employs modality-specific encoders and dual Q-Formers to distill language-relevant evidence from vision and touch into compact prefix tokens, which are projected into the embedding space of a large language model (LLM) to condition generation. To strengthen cross-modal grounding, we align each modality with text and explicitly couple vision and touch via a contrastive objective, promoting unified multi-property reasoning. We build **VitaSet**, a VTL dataset with 186 objects, 52k images across modalities, and 5.1k instruction–answer pairs with human-verified labels. Extensive experiments

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demonstrate that VitaTouch achieves state-of-the-art performance on the public TVL benchmark and delivers strong property understanding on VitaSet, achieving 88.89% hardness accuracy, 75.13% roughness accuracy, and 54.81% descriptor recall, together with a supplementary semantic similarity score of 0.9009. With parameter-efficient LoRA adaptation, VitaTouch maintains strong defect-recognition performance across increasing defect-category complexity, achieving 100.0%, 96.0%, and 92.0% accuracy for 2, 3, and 5 defect categories, respectively, with 50 labeled samples per category, and supports a proof-of-concept closed-loop robotic sorting workflow in a laboratory setting. More details are available at the project page: <https://vitatouch.github.io/>.

Keywords: Smart manufacturing, Defect recognition, Large language model, Vision–tactile sensing, Multimodal foundation model, Few-shot learning

1. Introduction

Quality inspection is a cornerstone of intelligent manufacturing, underpinning product reliability, production efficiency, and life-cycle cost control. In industrial practice, quality inspection should not be regarded as a single end-of-line decision; rather, it is a multi-layer activity that spans incoming material verification, process-aware monitoring during production, and final defect assessment before delivery. From the perspective of zero-defect manufacturing, automated in-line inspection enables real-time defect detection and adaptation to process disturbances, thereby providing essential support for quality control throughout multiple manufacturing stages [1]. In addition, robot-based intelligent inspection systems can achieve real-time defect identification, classification, and evaluation, while further enabling closed-loop adjustment of process parameters to suppress defect propagation before early degradation evolves into severe structural failure [2]. From an Industry 4.0 perspective, inspection strategies should also be continuously revised based on quality-cost considerations and production data so as to better balance conformance quality, production efficiency, and decision-making across the manufacturing lifecycle [3].

Despite rapid progress, mainstream industrial inspection pipelines remain predominantly vision-based. Modern machine-vision systems and deep models have achieved strong performance on a range of industrial benchmarks, es-

pecially under controlled conditions and with abundant training data. Large-scale anomaly detection datasets and surveys further consolidate the effectiveness of deep industrial anomaly detection (IAD) pipelines for image-level and pixel-level inspection [4, 5]. Meanwhile, systematic reviews of surface defect inspection using deep object detection networks highlight the maturity of vision-based solutions and their widespread adoption [6]. However, when quality inspection is viewed as multi-level physical understanding rather than appearance-based pattern recognition, purely visual pipelines exhibit intrinsic bottlenecks.

Vision alone provides an incomplete and fragile basis for physical quality assessment. It primarily captures geometry and photometric appearance, whereas many manufacturing-relevant cues are physical properties that are challenging to access in vision-only inspections, such as **hardness**, **roughness**, and **material**. Moreover, deployment in real shop-floor environments further amplifies failure modes: **occlusion** and visually inaccessible regions, **weak/uneven illumination**, and **specular reflections** on glossy or metallic surfaces can obscure defects and confound appearance cues. These issues are repeatedly observed as key pain points for deploying purely visual inspection in practice [6, 4]. Thus, improving reliability requires stepping beyond appearance-only cues toward sensing modalities that can directly capture the physical essence of manufactured surfaces.

Advances in tactile sensing and its growing adoption in robotic perception and manipulation provide a new paradigm for addressing the above limitations by introducing complementary physical evidence [7, 8]. Vision-based tactile sensors (VBTS) capture contact-induced deformation patterns that encode surface and material property cues while producing RGB tactile readouts. Compared with conventional vision, these signals provide richer local contact details than conventional vision and reduce sensitivity to illumination artifacts that often hinder visual imaging [8]. Recent progress in visuo-tactile representation learning further suggests the feasibility of learning transferable tactile foundations across sensors and data regimes [9]. These developments motivate property-centric inspection that is grounded in contact mechanics, enabling more direct access to the physical attributes that drive quality assessment and early defect discovery.

In parallel, foundation models and large language models (LLMs) have reshaped multimodal learning by enabling a unified semantic interface between low-level sensory inputs and high-level concepts. Vision-language pretraining (e.g., CLIP) establishes scalable alignment between images and language,

supporting open-vocabulary recognition and flexible prompting [10]. More recent multimodal LLM frameworks (e.g., BLIP-2, InstructBLIP, LLaVA) further enable instruction-following and open-ended generation conditioned on sensory evidence [11, 12, 13]. For inspection, this paradigm shift is particularly attractive because it moves beyond closed-set classification toward language-driven workflows, where operators can specify criteria in natural language and models can respond with open-form attribute descriptions and reasoning-based explanations when visual evidence is ambiguous. Meanwhile, early efforts have begun to incorporate touch into language-grounded perception, including touch–vision–language alignment via the TVL dataset and models [14], as well as tactile–language models for object property reasoning, such as Octopi [15]. However, these works are primarily developed for general robotic perception and everyday objects rather than industrial quality inspection, and they do not explicitly target the stringent requirements of manufacturing deployment. Consistent with recent benchmarks, current multimodal LLMs still fall short of industrial requirements for anomaly- and defect-related understanding [16]. More fundamentally, property-centric physical understanding grounded in tactile evidence has not yet been established as a reusable foundation for smart-manufacturing quality inspection.

Practical deployment further demands that such foundations be adaptable and deployable under industrial constraints. In many production lines, labeled defect data are limited and fragmented across product variants, and collecting representative defect samples is difficult due to low incidence rates and long-tail variability. Furthermore, industrial inspection systems often face strict requirements for inference latency and reliability, being deployed on industrial PCs or edge devices means limited GPU memory and compute budgets. These constraints make full-model finetuning impractical, especially when incorporating large multimodal models. Therefore, effective solutions must support few-shot adaptation with minimal labeled data, and must do so via resource-aware mechanisms that enable real-time inference and robust deployment on constrained hardware. Existing methods, however, typically lack property-aware multimodal grounding, which can render few-shot transfer brittle when appearance cues shift or when defects are primarily characterized by physical-property changes rather than visual patterns.

This paper argues that **multimodal physical-property understanding** is a missing foundation for reliable quality inspection. We propose **VitaTouch**, a **vision–tactile–language (VTL) foundation model** for embodied robotic quality inspection, designed to learn shared grounding among

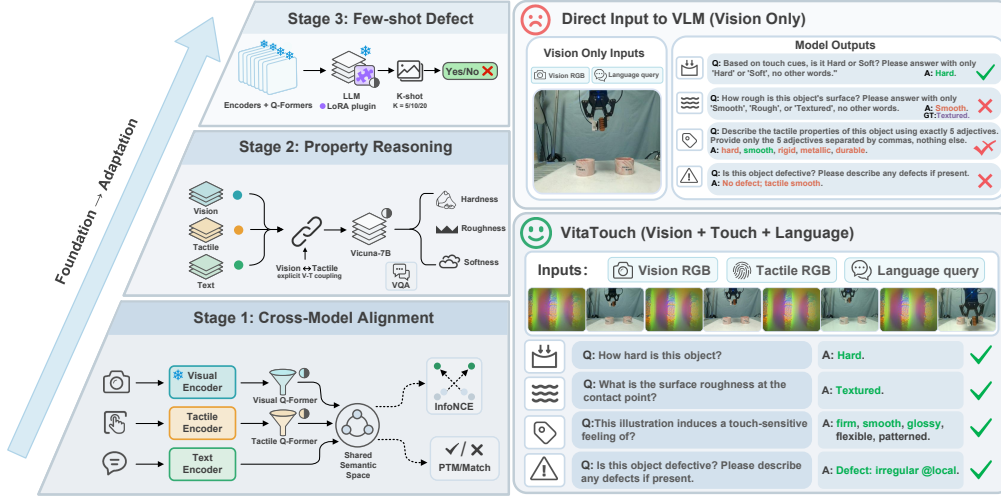


Figure 1: **Overview of VitaTouch.** Left: three-stage training pipeline. Stage 1: cross-modal alignment via dual Q-Formers (InfoNCE + PTM). Stage 2: property-reasoning foundation model with fused V-T tokens in frozen Vicuna-7B. LoRA-based defect adaptation across progressively finer-grained defect label spaces with few-shot labeled samples per category. Right: tactile sensing complements vision for property reasoning and defect recognition under visually ambiguous conditions.

visual observations, tactile evidence, and natural-language queries, thereby enabling multi-property inference and language-form outputs for material and surface understanding. VitaTouch follows a property-centric pretraining to deployment-oriented adaptation paradigm: it first establishes cross-modal grounding across vision, touch, and language; then learns reusable representations of material and surface properties; and finally supports efficient downstream adaptation to defect inspection under few-shot and resource-constrained settings via parameter-efficient updates (e.g., low-rank adaptation). By treating defect inspection as a downstream task built upon a general, property-aware foundation, VitaTouch better aligns with the multi-level nature of industrial quality inspection and enables effective transfer from property-centric understanding to practical defect recognition.

Contributions. Our main contributions are summarized as follows:

- We construct and release a curated vision-tactile-language (VTL) dataset for robotic manipulation. Built on our industrial subset and the Gel-Sight subset of AnyTouch under a unified annotation schema, it pro-

vides aligned visual, tactile, and language annotations for multimodal quality inspection.

- We propose a property-centric training pipeline that couples cross-modal alignment with multi-property attribute learning in the VTL setting, establishing a reusable backbone for material and surface property understanding.
- We introduce VitaTouch, a property-aware vision–tactile–language foundation model for multimodal quality inspection in smart manufacturing, targeting unified physical understanding rather than appearance-only detection.
- We demonstrate efficient transfer to few-shot defect recognition under resource constraints via parameter-efficient adaptation, validating that property-centric pretraining supports practical industrial inspection scenarios.

The rest of the article is organized as follows. Section 3 introduces VitaSet, the training and evaluation suite for industrial quality inspection. Section 4 presents the VitaTouch architecture and the progressive three-stage training protocol. Section 5 demonstrates experimental results on public benchmarks and VitaSet, together with ablation study, few-shot defect adaptation, and real-world sorting demonstrations. Finally, we conclude the paper with a discussion of limitations and future directions.

2. Related Work

2.1. Vision-Based Defect Inspection with Deep Learning

Vision-based inspection has long been the dominant paradigm in industrial quality inspection, and deep learning has substantially improved the accuracy and scalability of defect detection and localization. When dense labels are available, supervised detection and segmentation models provide strong performance in controlled settings. However, in many production lines, defective samples are rare, diverse, and costly to annotate, which has fueled the rapid development of industrial anomaly detection (IAD) approaches that rely primarily on normal data. Representative IAD lines include feature distribution modeling (e.g., PaDiM) [17], discriminative reconstruction and embedding learning (e.g., DRAEM) [18], and retrieval-based patch embedding

matching (e.g., PatchCore) [19]. Large-scale benchmarks such as MVTec AD [5] have become standard testbeds, while subsequent datasets aim to improve realism and coverage. Examples include VisA, introduced with the SPD pre-training framework [20], and Real-IAD for real-world multi-view inspection [21]. Comprehensive surveys and systematic reviews further summarize the maturity of deep IAD and surface defect inspection pipelines [4, 6]. Despite strong progress, purely visual pipelines remain sensitive to shop-floor factors such as occlusion, illumination changes, and specular reflections, and they are fundamentally limited when defects correlate more with deviations in physical property than stable appearance cues.

2.2. From Closed-Set Recognition to Language-Driven Inspection with LLM/VLMs

The recent surge of large language models (LLMs) and vision-language models (VLMs) is reshaping inspection, shifting from closed-set classification and detection to language-driven problem solving. Compared with conventional multi-class classifiers, VLM- and LLM-based systems offer (i) instruction following via natural-language queries, (ii) open-form descriptions and attribute reporting beyond a fixed label taxonomy, and (iii) semantic reasoning that can support interactive, operator-centric workflows. CLIP-style vision-language pretraining establishes scalable image-text alignment and enables promptable open-vocabulary recognition [10]. Building on this, multimodal LLM frameworks (e.g., BLIP-2, InstructBLIP, LLaVA) connect visual evidence to strong language decoders, enabling instruction tuning and generative responses conditioned on multimodal observations [11, 12, 13]. However, manufacturing inspection has domain-specific requirements (e.g., fine-grained localization, long-tail categories, safety-critical precision), and recent benchmarks suggest that even state-of-the-art multimodal LLMs still fall short for anomaly and defect-centric understanding under industrial conditions [16]. These observations motivate targeted research on leveraging language capabilities while respecting industrial constraints.

2.3. VLM/LLM for Defect and Anomaly Detection: Methods and Benchmarks

A fast-growing literature investigates how to adapt CLIP-style VLMs for anomaly and defect detection and segmentation, highlighting language as a scalable mechanism for generalization across product categories. A key challenge is that generic VLMs emphasize object semantics rather than normality or abnormality semantics, making naive prompting insufficient for anomaly

localization. WinCLIP addresses this issue with window- and patch-level feature aggregation and compositional prompt ensembling, enabling strong zero-/few-shot anomaly classification and segmentation on MVTec AD and VisA [22]. Prompt learning further reduces reliance on manual prompt engineering. For example, PromptAD proposes one-class prompt learning using only normal samples, improving few-shot anomaly detection under realistic supervision [23]. AnomalyCLIP learns object-agnostic prompts to capture generic normal/abnormal concepts and enhances cross-domain zero-shot anomaly detection [24]. Beyond prompt learning, context-conditioned prompting and cross-modal interactions have been explored to improve generalization. Representative examples include VCP-CLIP, which injects visual context into prompts to avoid product-specific prompt design [25], and AdaCLIP, which combines static and dynamic learnable prompts for zero-shot anomaly detection [26].

Recent work also composes multiple foundation models to improve localization quality. ClipSAM combines CLIP-style anomaly localization with SAM-based mask refinement to obtain sharper anomaly boundaries [27]. Meanwhile, anomaly-aware alignment has been studied to improve the discriminability of CLIP features for normal versus abnormal semantics: AA-CLIP proposes anomaly-aware text anchors and patch-level alignment while maintaining generalization [28], and FE-CLIP injects frequency cues into CLIP for improved zero-shot anomaly detection and segmentation [29]. Complementary to CLIP-centric pipelines, LLM-style industrial anomaly systems have emerged that emphasize dialogue and threshold-free outputs; for example, AnomalyGPT leverages large vision-language models with anomaly-oriented prompting and synthetic supervision to support multi-turn reporting and localization [30]. Collectively, methods evaluated on benchmarks such as MVTec AD [5], VisA [20], Real-IAD [21], and industrial MLLM benchmark MMAD [16], demonstrate that language can substantially enhance scalability, flexibility, and interpretability in defect and anomaly inspection.

2.4. Introducing Touch: Deep Visuo-Tactile Learning and LLM-Grounded Touch-Vision-Language

While language-guided VLMs improve semantic flexibility, reliable quality inspection also requires sensing modalities that capture physical surface and material cues beyond appearance. Vision-based tactile sensors (VBTS) provide image-like tactile observations that encode micro-geometry and contact

mechanics and are increasingly feasible in practical systems [8]. Representative hardware platforms include GelSight [31] and the compact DIGIT sensor [32]. On the learning side, tactile representation learning aims to build transferable encoders across sensors and tasks. For instance, AnyTouch studies unified static-dynamic representations across multiple visuo-tactile sensors [9]. In industrial contexts, multimodal inspection pipelines that combine global vision with local tactile scanning have shown practical value for detecting small defects on large-scale components, where a coarse-to-fine strategy mitigates coverage challenges and visual blind spots [33].

Importantly, touch research is now evolving from deep visuo-tactile fusion toward LLM-grounded tactile reasoning with natural language. The TVL dataset is a milestone, providing large-scale vision–touch pairs with language annotations and pseudo-labels, which enables alignment of tactile encoders with vision–language spaces and tactile-conditioned text generation [14]. Octopi further demonstrates tactile-language property reasoning by combining tactile representation learning with VLMs and exploiting intermediate property predictions (e.g., hardness, roughness) to improve downstream reasoning [15]. Nevertheless, these milestones are primarily developed for general robotics and everyday object understanding, and industrial inspection still lacks a reusable, property-centric touch–vision–language foundation designed around manufacturing defect taxonomies, process variability, and deployment constraints.

2.5. Few-Shot Adaptation and Parameter-Efficient Tuning for Industrial Deployment

Deployment constraints for industrial quality inspection differ from general-purpose multimodal reasoning in several respects. **First**, defect data are scarce: failure modes are rare, diverse across product lines, and expensive to label due to domain-expertise requirements. **Second**, production environments are dynamic: new variants and process drift emerge continuously, requiring rapid model updates without exhaustive retraining. **Third**, computational budgets are tight: on-site inference runs on edge devices (e.g., industrial PCs) under strict latency and memory constraints, making deployment of large multimodal models challenging.

To address these constraints, different learning paradigms have been proposed. For data and task constraints, few-shot learning and meta-learning aim for rapid adaptation to new tasks using limited examples; representative methods include MAML [34] and prototypical networks [35]. For resource

constraints, parameter-efficient tuning provides a practical path to adapting foundation models under compute and memory limits by updating only small trainable modules while keeping backbones frozen. LoRA, for example, injects trainable low-rank parameters into pretrained models and significantly reduces adaptation cost [36]. Most few-shot anomaly inspection approaches remain vision-centric and thus lack property-aware multimodal grounding. When defects correspond to physical property deviations rather than stable visual patterns, models without tactile evidence and explicit property understanding can become brittle under distribution shifts, motivating multimodal property-centric pretraining to support robust few-shot transfer in manufacturing.

2.6. Positioning

To summarize, prior work has (i) established strong vision-based IAD methods and benchmarks [5, 19, 18, 17, 20, 21, 4, 6]; (ii) advanced language-guided anomaly/defect detection by adapting CLIP/VLMs via prompting, prompt learning, context injection, and foundation-model composition [22, 23, 24, 25, 26, 27, 28, 29]; and (iii) introduced touch sensing, tactile representation learning, and emerging touch–vision–language milestones that enable tactile-grounded language interaction and reasoning [8, 31, 32, 9, 33, 14, 15]. These trends motivate building a property-aware vision–tactile–language foundation for industrial inspection that unifies open-language interaction, physically grounded perception, and resource-aware adaptation to few-shot defect tasks.

3. Dataset and Training & Evaluation Suite

This section introduces **VitaSet**, a vision–tactile–language (VTL) dataset and training/evaluation suite designed for industrial quality inspection. Existing visuo-tactile resources are mostly collected around everyday objects and general robotic perception, and they provide limited coverage of industrial operating conditions and inspection-oriented supervision. To bridge this gap, we build VitaSet as a **curated fusion dataset** that combines an industrial-scene robotic manipulation subset collected on our platform with the GelSight-only subset of AnyTouch. Under a unified annotation schema, VitaSet supports property-centric pretraining and inspection-oriented evaluation for our VitaTouch model, while the released dataset is intended to

support broader multimodal research beyond the specific tasks studied in this paper.

3.1. Physical Property Selection

Industrial quality inspection verifies whether products conform to physical specifications that determine functionality and process stability. We therefore focus on three inspection-relevant properties that are (i) well-defined and measurable, (ii) frequently implicated in industrial nonconformities, and (iii) naturally supported by contact sensing in addition to RGB appearance.

Hardness. Hardness quantifies resistance to indentation/deformation under applied pressure. In manufacturing workflows, hardness is a standard incoming-inspection attribute and an in-process conformity indicator; deviations can signal material mismatch or process drift even when RGB appearance remains inconclusive.

Roughness. Roughness characterizes surface micro-geometry that governs friction, wear, sealing, and coating adhesion. In shop-floor conditions, purely visual roughness cues can be unstable due to illumination variation, specular reflections, and low-texture surfaces. GelSight contact images provide direct evidence of micro-geometry changes under contact, complementing RGB observations.

Material category. Material category constrains expected mechanical and tribological behavior and is central to inspection procedures such as incoming checks, mixed-material prevention, and production-line sorting. VitaSet operationalizes material-related supervision in an inspection-reporting style: the model is asked to output a fixed-length list of tactile descriptors (e.g., top-5 adjectives from a controlled vocabulary) that best match the perceived material/surface feel, which enables auditable and actionable reporting for quality-inspection decisions.

3.2. Dataset Construction and Annotation

We release a dataset

$$\mathcal{D} = \{(\mathbf{X}_i, T_i, Y_i)\}_{i=1}^N, \quad \mathbf{X}_i = (\mathbf{X}_i^v, \mathbf{X}_i^t), \quad (1)$$

where $\mathbf{X}_i^v$ denotes the RGB image, $\mathbf{X}_i^t$ denotes the GelSight tactile image, T_i is the instruction/query, and Y_i is the corresponding text-form answer. We cast both discrete labels and descriptive outputs into text-form answers, enabling unified instruction-following supervision across diverse tasks.

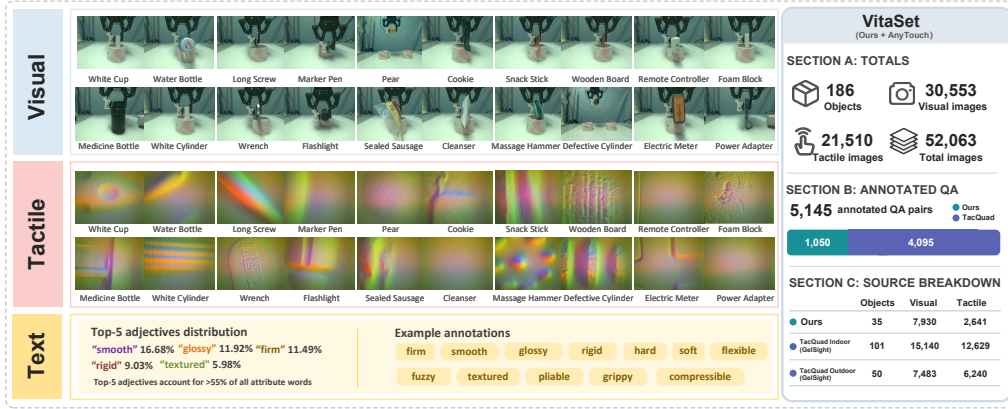


Figure 2: **VitaSet overview (Ours + AnyTouch GelSight-only)**. Aligned RGB observations and paired GelSight tactile readings across objects, with controlled-vocabulary annotations and dataset statistics under a unified schema.

VitaSet integrates two complementary data sources to improve diversity and coverage. First, we collect an industrial-scene robotic manipulation dataset using a robotic platform equipped with an RGB camera and a GelSight tactile sensor; data are captured under repeated grasp-and-contact interactions that mimic real-world inspection-like operating conditions. Second, we incorporate the GelSight-only subset of the AnyTouch dataset. Since AnyTouch contains multiple sensor modalities, we retain only its GelSight portion to ensure cross-source modality consistency. This integration increases the diversity of object instances and contact outcomes while maintaining tactile modality alignment for unified model training.

In total, VitaSet comprises **186 objects**, including **30,553** RGB images and **21,510** GelSight tactile images, for a total of **52,063** images across modalities. We further provide **5,145** annotated instruction-answer (QA) pairs. The smaller number of tactile images is primarily due to the manual removal of a substantial number of invalid tactile frames that do not correspond to valid object contact (e.g., no-contact or low-quality contact instances), while the corresponding RGB stream is retained for all trials.

To ensure high-quality and consistent annotations, we adopt a scalable yet auditable annotation pipeline. Initial instruction-answer pairs are first generated via the **GPT-4o** API under template-constrained prompting: classification tasks are restricted to a predefined label set, while description tasks are bounded to a controlled vocabulary with a fixed-length format.

These generated pairs are not directly used as final annotations. Instead, human annotators conduct full object-level reviews of all generated labels across both our in-house collected data and the integrated AnyTouch subset. During this verification step, annotators correct inconsistent, implausible, or format-violating outputs, thereby mitigating model hallucinations and ensuring a unified and consistent label space across the entire dataset. This full-coverage verification, conducted over all **186 objects** rather than partial sampling, yields an auditable annotation schema that aligns both data sources. An overview of VitaSet, including examples from both sources and the unified annotation schema, is shown in Fig. 2.

3.3. Training and Evaluation Suite

We define a suite of tasks aligned with quality-inspection workflows. All tasks are cast into an instruction-following form $(\mathbf{X}, T) \rightarrow Y$, where $\mathbf{X} = (\mathbf{X}^v, \mathbf{X}^t)$ denotes the multimodal input consisting of an RGB image and a tactile image, T is the instruction/query, and Y is the target response.

Task 1: Hardness classification (binary). Given $\mathbf{X}$ and a hardness query, the model outputs one token from $\{\text{Hard}, \text{Soft}\}$. **Metric:** Accuracy. We additionally report F1 when class imbalance is non-negligible.

Task 2: Roughness classification (3-way). Given $\mathbf{X}$ and a roughness query, the model outputs one token from $\{\text{Smooth}, \text{Textured}, \text{Rough}\}$. **Metric:** Accuracy. When necessary, we further analyze class-wise performance for adjacent-category ambiguity.

Task 3: Material description. Given $\mathbf{X}$ and a material-related query, the model predicts a textual answer describing the material/surface properties. **Metric:** We evaluate this task using two complementary metrics: descriptor recall under strict lexical matching, which measures the proportion of ground-truth descriptors exactly recovered by the prediction, and a supplementary semantic-similarity score between the predicted and ground-truth texts.

Task 4: Defect recognition. Given $\mathbf{X}$ and a defect-related instruction, the model predicts a text-form label from a predefined defect taxonomy. We consider a progressively finer-grained evaluation protocol with 2-, 3-, and 5-category settings, corresponding to defect presence recognition, defect-type recognition, and joint defect-type/location recognition, respectively. **Metric:** Accuracy for category recognition. Short defect descriptions are evaluated qualitatively using inspection-style examples.

Overall, this suite couples property-centric physical understanding (Tasks 1–3) with inspection-oriented defect reporting (Task 4), supporting both

foundation-model training and downstream adaptation under practical inspection settings.

4. Method

VitaTouch is a vision–tactile–language model for property reasoning and few-shot defect inspection. As shown in Fig. 1, we train VitaTouch with a progressive three-stage protocol: (i) cross-modal alignment to establish a shared semantic interface across vision, touch, and language; (ii) property-reasoning foundation learning by conditioning a frozen LLM on fused vision–tactile tokens; and (iii) parameter-efficient few-shot adaptation to defect recognition using LoRA. The overall training procedure is summarized in the pseudo-code at the end of this section.

4.1. Problem Formulation

Each sample consists of an RGB image, a tactile image, and a natural-language query,

$$\mathcal{X}_i = (\mathbf{X}_i^v, \mathbf{X}_i^t, T_i), \quad (2)$$

where $\mathbf{X}_i^v \in \mathbb{R}^{H_v \times W_v \times 3}$ denotes the visual observation, $\mathbf{X}_i^t \in \mathbb{R}^{H_t \times W_t \times 3}$ denotes the tactile observation, and T_i specifies either a property query or a defect query. Given $\mathcal{X}_i$, VitaTouch generates a textual response Y_i .

4.2. Model Architecture

An overview of the proposed architecture is provided in Fig. 3. VitaTouch follows a dual-branch design with modality-specific encoders and Q-Formers. The Q-Formers distill compact, language-relevant tokens from vision and touch, which are then mapped into the LLM embedding space as prefix tokens.

4.2.1. Perception Encoders

Given $\mathbf{X}_i^v$, the visual encoder produces a token sequence

$$\mathbf{E}_i^v = f_\theta^v(\mathbf{X}_i^v) \in \mathbb{R}^{L_v \times d_v}. \quad (3)$$

Given $\mathbf{X}_i^t$, the tactile encoder produces

$$\mathbf{E}_i^t = f_\theta^t(\mathbf{X}_i^t) \in \mathbb{R}^{L_t \times d_t}. \quad (4)$$

The visual encoder is frozen throughout all stages. The tactile encoder is frozen in Stage 1 and fully fine-tuned in Stage 2, while it is kept frozen in Stage 3.

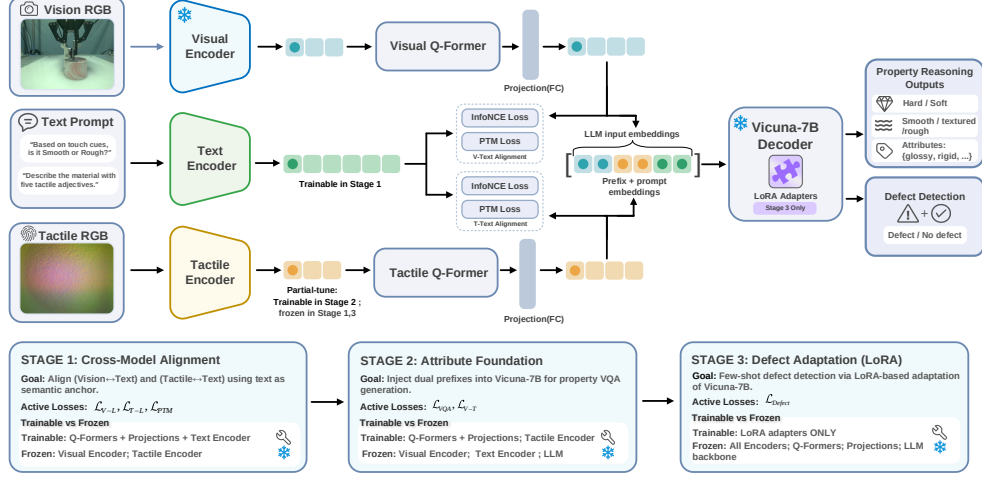


Figure 3: **VitaTouch model architecture.** VitaTouch employs a **dual-branch vision-tactile design** with modality-specific encoders and Q-Formers. The Q-Formers distill learnable queries into **vision-tactile prefix tokens**, which are prepended to text embeddings and fed into a frozen Vicuna-7B decoder. Training proceeds in three stages: **Stage 1** aligns cross-modal embeddings via frozen encoders; **Stage 2** establishes the perception-to-language pathway for property reasoning; **Stage 3** adapts the model for few-shot defect recognition using LoRA while freezing the backbone.

4.2.2. Dual Q-Former

For each modality $m \in \{v, t\}$, we instantiate an independent Q-Former with L_q learnable query vectors,

$$\mathbf{Q}_0^m \in \mathbb{R}^{L_q \times d_q}, \quad L_q = 32. \quad (5)$$

Given perception tokens $\mathbf{E}_i^m$, the Q-Former outputs

$$\mathbf{Q}_i^m = \text{QF}^m(\mathbf{Q}_0^m, \mathbf{E}_i^m) \in \mathbb{R}^{L_q \times d_q}. \quad (6)$$

Each query token is projected into a shared embedding space and ℓ_2 -normalized:

$$\hat{\mathbf{q}}_{ik}^m = \frac{\mathbf{W}_p^m \mathbf{Q}_{ik}^m}{\|\mathbf{W}_p^m \mathbf{Q}_{ik}^m\|_2}, \quad \hat{\mathbf{q}}_{ik}^m \in \mathbb{R}^d, \quad k = 1, \dots, L_q. \quad (7)$$

4.2.3. Text Encoding and LLM Interface

Given query text T_i , a text encoder produces token states $\mathbf{Z}_i$. We project the [CLS] state into the shared embedding space:

$$\hat{\mathbf{t}}_i = \frac{\mathbf{W}_t \mathbf{Z}_{i, [\text{CLS}]}}{\|\mathbf{W}_t \mathbf{Z}_{i, [\text{CLS}]}\|_2} \in \mathbb{R}^d. \quad (8)$$

For Stage 2/3, query tokens are mapped into the LLM embedding space:

$$\tilde{\mathbf{Q}}_i^v, \tilde{\mathbf{Q}}_i^t \in \mathbb{R}^{L_q \times d_{\text{LLM}}}, \quad \tilde{\mathbf{Q}}_i^{vt} = [\tilde{\mathbf{Q}}_i^v; \tilde{\mathbf{Q}}_i^t] \in \mathbb{R}^{2L_q \times d_{\text{LLM}}}. \quad (9)$$

4.3. Stage 1: Cross-Modal Alignment Pretraining

Stage 1 learns a shared semantic interface by aligning each modality with language. For a minibatch of size N , we pool the query set by averaging:

$$\hat{\mathbf{z}}_i^m = \frac{1}{L_q} \sum_{k=1}^{L_q} \hat{\mathbf{q}}_{ik}^m \in \mathbb{R}^d, \quad m \in \{v, t\}. \quad (10)$$

We apply symmetric bidirectional InfoNCE between modality m and language:

$$\mathcal{L}_{m-L} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{\exp(\langle \hat{\mathbf{z}}_i^m, \hat{\mathbf{t}}_i \rangle / \tau)}{\sum_{j=1}^N \exp(\langle \hat{\mathbf{z}}_i^m, \hat{\mathbf{t}}_j \rangle / \tau)} + \log \frac{\exp(\langle \hat{\mathbf{t}}_i, \hat{\mathbf{z}}_i^m \rangle / \tau)}{\sum_{j=1}^N \exp(\langle \hat{\mathbf{t}}_i, \hat{\mathbf{z}}_j^m \rangle / \tau)} \right] \quad (11)$$

The overall contrastive loss is defined as the sum of the modality–language contrastive terms:

$$\mathcal{L}_{\text{con}} = \mathcal{L}_{v-L} + \mathcal{L}_{t-L}. \quad (12)$$

To encourage semantic compatibility beyond metric alignment, we use perceptual-text matching with hard negatives:

$$\mathcal{L}_{\text{match}} = -\frac{1}{M} \sum_{\ell=1}^M \left[y_\ell \log p_\ell + (1 - y_\ell) \log(1 - p_\ell) \right], \quad (13)$$

The Stage 1 objective is

$$\mathcal{L}_{\text{Stage1}} = \mathcal{L}_{\text{con}} + \lambda_{\text{match}} \mathcal{L}_{\text{match}}. \quad (14)$$

4.4. Stage 2: Property-Reasoning Foundation Model

Stage 2 trains the model to answer property-related queries conditioned on fused vision–tactile prefixes. Given a property prompt T_i^{prop} and target sequence Y_i , we minimize the autoregressive loss:

$$\mathcal{L}_{\text{VQA}} = -\frac{1}{N} \sum_{i=1}^N \sum_{s=1}^{L_i} \log p_\theta(y_{i,s} \mid y_{i,<s}, \mathbf{X}_i^v, \mathbf{X}_i^t, T_i^{\text{prop}}). \quad (15)$$

We further enforce explicit vision–tactile coupling with a symmetric InfoNCE loss computed between the pooled vision and tactile embeddings:

$$\mathcal{L}_{v-t} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{\exp(\langle \hat{\mathbf{z}}_i^v, \hat{\mathbf{z}}_i^t \rangle / \tau_{vt})}{\sum_{j=1}^N \exp(\langle \hat{\mathbf{z}}_i^v, \hat{\mathbf{z}}_j^t \rangle / \tau_{vt})} + \log \frac{\exp(\langle \hat{\mathbf{z}}_i^t, \hat{\mathbf{z}}_i^v \rangle / \tau_{vt})}{\sum_{j=1}^N \exp(\langle \hat{\mathbf{z}}_i^t, \hat{\mathbf{z}}_j^v \rangle / \tau_{vt})} \right], \quad (16)$$

where τ_{vt} is a temperature hyperparameter.

The Stage 2 objective is

$$\mathcal{L}_{\text{Stage2}} = \mathcal{L}_{\text{VQA}} + \lambda_{\text{VT}} \mathcal{L}_{v-t}. \quad (17)$$

4.5. Stage 3: LoRA-Based Defect Adaptation

Stage 3 adapts the Stage 2 foundation model to **defect recognition** under data-limited settings. To enable parameter-efficient transfer, we freeze the perception encoders, Q-Formers, and modality-to-LLM projection layers, and insert LoRA modules into the attention projections of the Vicuna decoder. Specifically, the adapted weight is written as

$$\mathbf{W}' = \mathbf{W} + \frac{\alpha}{r} \mathbf{B} \mathbf{A}, \quad (18)$$

where r is the LoRA rank and α is the scaling factor.

Given a defect-related prompt T_i^{def} and a target answer sequence $Y_i = (y_{i,1}, \dots, y_{i,L_i})$, we optimize the standard autoregressive cross-entropy loss:

$$\mathcal{L}_{\text{defect}} = -\frac{1}{N} \sum_{i=1}^N \sum_{s=1}^{L_i} \log p_{\theta}(y_{i,s} \mid y_{i,<s}, \mathbf{X}_i^v, \mathbf{X}_i^t, T_i^{\text{def}}), \quad (19)$$

where L_i denotes the token length of the target answer sequence Y_i , and $y_{i,<s} = (y_{i,1}, \dots, y_{i,s-1})$ denotes the target prefix before step s . In our setting, the target answer sequence typically corresponds to a defect-category label or **Normal** in text form.

At inference time, the model generates a textual response conditioned on the multimodal input and the defect prompt:

$$\hat{Y}_i \sim p_{\theta}(\cdot \mid \mathbf{X}_i^v, \mathbf{X}_i^t, T_i^{\text{def}}). \quad (20)$$

The generated response is then mapped to the predefined coarse-to-fine defect taxonomy for evaluation. In this way, Stage 3 preserves the language-form prediction interface of VitaTouch while enabling efficient downstream adaptation for defect recognition.

Overall training procedure. For completeness, Algorithm 1 summarizes our three-stage training pipeline. Stage 1 learns language-anchored cross-modal alignment between vision, touch, and text (InfoNCE + PTM). Stage 2 trains the property-reasoning foundation model by conditioning a frozen LLM on fused vision–tactile prefix tokens and optimizing an instruction-following objective with an additional vision–tactile coupling loss. Stage 3 performs parameter-efficient adaptation to defect recognition by inserting LoRA modules into the LLM and updating only the LoRA parameters.

5. Experiments

Our experiments follow a progressive validation pipeline from public benchmarking to manufacturing-oriented evaluation and deployment. We first evaluate VitaTouch on the public TVL benchmark to assess its overall capability and enable direct comparison with prior methods. We then report multi-task property understanding on our curated VitaSet dataset tailored for manufacturing-relevant attributes, followed by an ablation study to analyze the contribution of key components. Next, we evaluate parameter-efficient few-shot defect recognition via LoRA adaptation under limited labels. We further provide qualitative inspection-style outputs to illustrate format compliance and interpretability. Finally, we demonstrate a closed-loop robotic sorting system to verify that model predictions are actionable in a physical quality-inspection workflow.

5.1. Datasets and Benchmarks

TVL benchmark. We evaluate VitaTouch on the public Touch–Vision–Language (TVL) benchmark [14]. Following the official protocol, we use the original benchmark setup and data split, without introducing additional external data. Results are reported on the Self-Supervised Visuo-Tactile Pretraining (SSVTP) subset, the Human Collected Tactile (HCT) subset, and the overall TVL benchmark for direct comparison with prior methods.

Curated VitaSet dataset. To better reflect smart manufacturing inspection requirements, we collect a synchronized vision–tactile–language dataset with an RGB camera and a GelSight tactile sensor. The dataset targets manufacturing-relevant physical understanding and supports the following tasks: (i) hardness perception (binary), (ii) roughness recognition (3-way), and (iii) material property description (tactile attribute generation). We also support

language-form outputs for operator-facing reporting (controlled-vocabulary attributes / short descriptions), consistent with the Stage 2 instruction-following formulation.

5.2. Implementation Details

VitaTouch is implemented in PyTorch. We use EVA-CLIP-G as the visual encoder and an AnyTouch encoder (ViT-L backbone) for tactile inputs. The language backbone is Vicuna-7B.

Stage 1: cross-modal alignment. Stage 1 performs explicit cross-modal alignment. We train only the two modality-specific Q-Formers (vision and tactile) together with their projection layers that map query tokens into a shared embedding space, while keeping the perception encoders fixed. We use AdamW with a linear-warmup cosine learning-rate schedule, using a peak learning rate of 1×10^{-4} , warmup learning rate of 1×10^{-6} , 500 warmup steps, and weight decay of 0.01. Automatic mixed precision (AMP) is enabled throughout training.

Stage 2: property-reasoning foundation learning. Stage 2 performs multi-task instruction supervision for property understanding. In this stage, we freeze the LLM (Vicuna-7B) and the visual encoder, and finetune the tactile encoder, both Q-Formers, and the modality-to-LLM projection layers end-to-end to build a property-aware foundation model. We use AdamW with the same linear-warmup cosine schedule (peak LR 1×10^{-4} , warmup LR 1×10^{-6} , 500 warmup steps, weight decay 0.01) and AMP for stable large-model training. We set the maximum input/output length to 512 tokens to support inspection-style instruction outputs.

Stage 3: LoRA-Based Defect Adaptation. Stage 3 adapts the Stage 2 foundation model to defect recognition under limited labels. We freeze the perception encoders, Q-Formers, and projection layers, and apply LoRA to the Vicuna attention projections, updating only the LoRA parameters. We adopt a LoRA configuration with rank $r = 16$, scaling factor $\alpha = 32$, and dropout rate of 0.1 for efficient adaptation.

Training hardware and time. Stage 1 and Stage 2 are trained on an 8×NVIDIA A800 GPU cluster, taking approximately 2 hours (Stage 1) and 10 hours (Stage 2), respectively. Stage 3 is lightweight and can be completed within 1 hour on a single RTX 5090 GPU, which aligns with practical industrial deployment constraints.

Table 1: Comparison on the TVL benchmark. VitaTouch achieves the best performance on HCT and TVL, and remains competitive on SSVTP.

Model	Encoder Pre-training Modalities			Score (1–10)		
	Vision	Tactile	Language	SSVTP	HCT	TVL
LLaVA-1.5 7B	✓	-	✓	3.64	3.55	3.56
LLaVA-1.5 13B	✓	-	✓	3.55	3.63	3.62
ViP-LLaVA 7B	✓	-	✓	2.72	3.44	3.36
ViP-LLaVA 13B	✓	-	✓	4.10	3.76	3.80
LLaMA-Adapter	✓	-	✓	2.56	3.08	3.02
BLIP-2 Opt-6.7b	✓	-	✓	2.02	2.72	2.64
InstructBLIP 7B	✓	-	✓	1.40	1.30	1.31
InstructBLIP 13B	✓	-	✓	1.44	1.21	1.24
GPT-4V	✓	-	✓	5.02	4.42	4.49
SSVTP-LLaMA	✓	✓	-	2.58	3.67	3.54
TVL-LLaMA (ViT-Tiny)	✓	✓	✓	6.09	4.79	4.94
TVL-LLaMA (ViT-Small)	✓	✓	✓	5.81	4.77	4.89
TVL-LLaMA (ViT-Base)	✓	✓	✓	6.16	4.89	5.03
VitaTouch (Ours)	✓	✓	✓	5.83	6.10	6.09

5.3. Main Results

5.3.1. Results on the TVL Benchmark

We first evaluate VitaTouch on the public Touch–Vision–Language (TVL) benchmark [14] to assess its capability under a standardized evaluation protocol and to enable direct comparison with prior methods. The benchmark includes the Self-Supervised Visuo-Tactile Pretraining (SSVTP) subset and the Human Collected Tactile (HCT) subset, as defined in the TVL benchmark [14]. Table 1 shows that VitaTouch achieves the best scores on **HCT** and **TVL** among the compared methods, while remaining competitive on **SSVTP**.

Algorithm 1 VitaTouch training procedure (three-stage pipeline)

Require: training set $\mathcal{D} = \{(\mathbf{X}_i^v, \mathbf{X}_i^t, T_i)\}$; property prompts T_i^{prop} with targets Y_i ; defect prompts T_i^{def} with target sequences Y_i

- 1: Initialize learnable queries $\mathbf{Q}_0^v, \mathbf{Q}_0^t$ and projection layers
- 2: Freeze visual encoder f_θ^v throughout training and freeze Vicuna-7B

3: **Stage 1: Cross-modal alignment (InfoNCE + PTM)**

- 4: **for** each minibatch $\{(\mathbf{X}_i^v, \mathbf{X}_i^t, T_i)\}_{i=1}^N$ **do**
- 5: Freeze tactile encoder f_θ^t
- 6: $\mathbf{E}_i^v \leftarrow f_\theta^v(\mathbf{X}_i^v)$; $\mathbf{E}_i^t \leftarrow f_\theta^t(\mathbf{X}_i^t)$
- 7: $\mathbf{Q}_i^m \leftarrow \text{QF}^m(\mathbf{Q}_0^m, \mathbf{E}_i^m)$; $\hat{\mathbf{q}}_{ik}^m \leftarrow \ell_2\text{-norm}(\mathbf{W}_p^m \mathbf{Q}_{ik}^m)$, $m \in \{v, t\}$
- 8: $\mathbf{Z}_i \leftarrow \text{Enc}^{\text{text}}(T_i)$; $\hat{\mathbf{t}}_i \leftarrow \ell_2\text{-norm}(\mathbf{W}_t \mathbf{Z}_{i, [\text{CLS}]})$
- 9: Compute pooled embeddings $\hat{\mathbf{z}}_i^m \leftarrow \frac{1}{L_q} \sum_{k=1}^{L_q} \hat{\mathbf{q}}_{ik}^m$ (Eq. (10))
- 10: $\mathcal{L}_{\text{con}} \leftarrow \mathcal{L}_{v-L} + \mathcal{L}_{t-L}$ (Eq. (12))
- 11: $\mathcal{L}_{\text{match}} \leftarrow \text{Eq. (13)}$
- 12: $\mathcal{L}_{\text{Stage1}} \leftarrow \mathcal{L}_{\text{con}} + \lambda_{\text{match}} \mathcal{L}_{\text{match}}$
- 13: Update Stage 1 trainable parameters by $\nabla \mathcal{L}_{\text{Stage1}}$
- 14: **end for**

15: **Stage 2: Property-aware instruction learning**

- 16: Unfreeze and fine-tune tactile encoder f_θ^t in Stage 2; keep f_θ^v and Vicuna-7B frozen
- 17: **for** each minibatch $\{(\mathbf{X}_i^v, \mathbf{X}_i^t, T_i^{\text{prop}}, Y_i)\}_{i=1}^N$ **do**
- 18: Compute prefixes $\tilde{\mathbf{Q}}_i^{\text{vt}} = [\tilde{\mathbf{Q}}_i^v; \tilde{\mathbf{Q}}_i^t]$ and prepend them to the text embeddings of T_i^{prop}
- 19: $\mathcal{L}_{\text{VQA}} \leftarrow \text{Eq. (15)}$
- 20: $\mathcal{L}_{v-t} \leftarrow \text{Eq. (16)}$
- 21: $\mathcal{L}_{\text{Stage2}} \leftarrow \mathcal{L}_{\text{VQA}} + \lambda_{\text{VT}} \mathcal{L}_{v-t}$
- 22: Update Stage 2 trainable parameters by $\nabla \mathcal{L}_{\text{Stage2}}$
- 23: **end for**

24: **Stage 3: Few-shot defect adaptation (LoRA)**

- 25: Insert LoRA into Vicuna attention projections; freeze f_θ^v , f_θ^t , Q-Formers, and projection layers
 - 26: **for** each minibatch $\{(\mathbf{X}_i^v, \mathbf{X}_i^t, T_i^{\text{def}}, Y_i)\}_{i=1}^N$ **do**
 - 27: Condition the frozen backbone on $(\tilde{\mathbf{Q}}_i^{\text{vt}}, T_i^{\text{def}})$
 - 28: $\mathcal{L}_{\text{defect}} \leftarrow \text{Eq. (19)}$
 - 29: Update LoRA parameters only
 - 30: **end for**
-

5.3.2. Property Understanding on the VitaSet Dataset

To evaluate VitaTouch for manufacturing-oriented quality inspection, we conduct experiments on the curated VitaSet dataset. The evaluation includes three property-related tasks: hardness classification, roughness classification, and material descriptor prediction. Hardness and roughness are treated as classification tasks and evaluated by accuracy, whereas the material task is formulated as descriptor prediction, where the model generates a set of material-related descriptors rather than a single class label.

Unless otherwise stated, the main quantitative results reported in this subsection use the checkpoint selected by the highest mean task score on the validation set.

For the material descriptor task, let G_i denote the set of five human-annotated reference descriptors for sample i , and let P_i denote the set of valid predicted descriptors after duplicate removal and invalid-word filtering. We first evaluate strict lexical matching using descriptor recall:

$$R_{\text{desc}} = \frac{1}{N} \sum_{i=1}^N \frac{|P_i \cap G_i|}{|G_i|}, \quad (21)$$

where N is the number of validation samples. This metric measures the average proportion of reference descriptors exactly recovered by the model.

Because strict lexical matching does not reward semantically similar alternatives, we further introduce a supplementary semantic-similarity evaluation. Let y_i^{pred} and y_i^{gt} denote the predicted and ground-truth descriptor texts, respectively. Using Qwen3-Embedding-0.6B, we obtain normalized sentence embeddings

$$\mathbf{z}_i^{\text{pred}} = \frac{E_\phi(y_i^{\text{pred}})}{\|E_\phi(y_i^{\text{pred}})\|_2}, \quad \mathbf{z}_i^{\text{gt}} = \frac{E_\phi(y_i^{\text{gt}})}{\|E_\phi(y_i^{\text{gt}})\|_2}, \quad (22)$$

where $E_\phi(\cdot)$ denotes the sentence embedding extracted from the last valid token in the final hidden state of Qwen3-Embedding-0.6B. The semantic similarity is then defined as

$$S_{\text{sem}} = \frac{1}{N} \sum_{i=1}^N (\mathbf{z}_i^{\text{pred}})^\top \mathbf{z}_i^{\text{gt}}, \quad (23)$$

which is equivalent to the mean cosine similarity over the validation set.

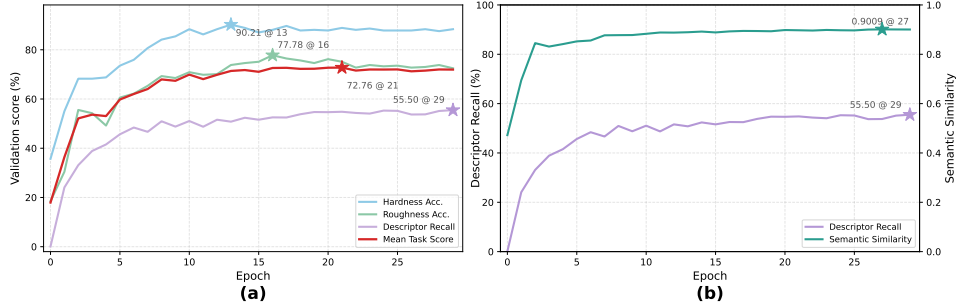


Figure 4: VitaSet validation performance of VitaTouch across training epochs. (a) Multi-task validation trends for hardness accuracy, roughness accuracy, descriptor recall, and mean task score. (b) Comparison of strict exact-match descriptor recall and semantic similarity for the material descriptor task. Stars (★) mark the best epoch for each metric.

Figure 4 summarizes the validation trends across training epochs. Since different tasks reach their best values at different epochs, we use the checkpoint with the highest mean task score on the validation set as the main checkpoint for quantitative reporting. Specifically, the selected main checkpoint is epoch 21, which achieves 88.89% hardness accuracy, 75.13% roughness accuracy, and 54.81% descriptor recall, with a mean task score of 72.76%. For completeness, Fig. 4(a) also marks the best epoch reached by each individual metric. The best single-task values observed during training are 90.21% hardness accuracy at epoch 13, 77.78% roughness accuracy at epoch 16, and 55.50% descriptor recall at epoch 29. These task-specific peak values are shown only for analyzing training dynamics, and Table 2 presents the results of the single main checkpoint determined by the highest mean task score.

Fig. 4(b) focuses on the material descriptor task: despite the strict exact-match criterion of descriptor recall, the semantic similarity remains high and reaches 0.9009 at epoch 27, verifying that VitaTouch generates semantically aligned descriptors even without exact lexical matches, which supports its practical value in engineering inspection reporting.

5.3.3. Ablation Study

Table 2 and Fig. 5 summarize the ablation results on VitaSet. Using the main checkpoint selected by the highest mean task score, the full VitaTouch model achieves the best overall performance, reaching 88.89% hardness accuracy, 75.13% roughness accuracy, and 54.81% descriptor recall, with a

Table 2: Comparison of the full VitaTouch model, a no-Stage-1 variant, and unimodal settings on VitaSet across tasks.

Method	Hardness Acc.	Roughness Acc.	Descriptor Recall	Mean Task Score
Vision only (Qwen-2.5-VL)	73.81%	61.11%	32.28%	55.73%
Tactile only (Octopi)	68.25%	58.20%	8.40%	44.95%
VitaTouch (Without Stage 1 alignment)	84.80%	72.50%	17.46%	57.18%
VitaTouch (Ours, main checkpoint)	88.89%	75.13%	54.81%	72.76%

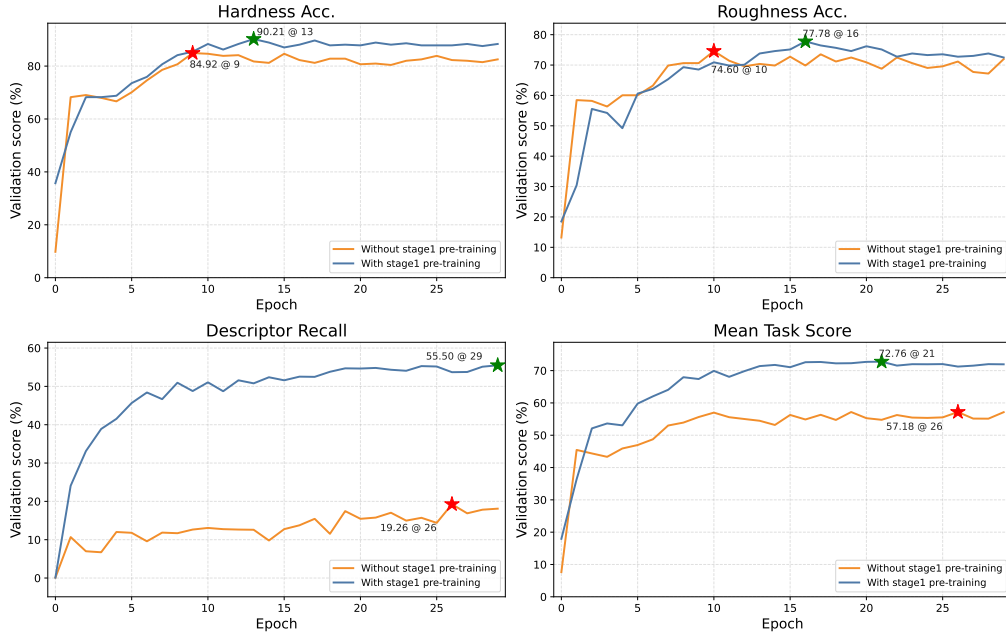


Figure 5: Ablation results on the VitaSet dataset across tasks. Each variant removes one key stage from the full model, demonstrating the necessity of explicit alignment and multimodal fusion for robust multi-task property learning.

mean task score of 72.76%. It is worth noting that descriptor recall is a stricter metric than classification accuracy, since it requires recovering multiple human-annotated descriptors exactly rather than predicting a single label.

Limitations of unimodal approaches. Both unimodal baselines perform substantially worse than the full model. The vision-only baseline achieves 73.81% hardness accuracy and 61.11% roughness accuracy, but only 32.28% descriptor recall, suggesting that appearance cues alone are insufficient for reliable material descriptor prediction. The tactile-only baseline performs even worse on the descriptor task, achieving only 8.40% descriptor recall. This result indicates that localized contact information without explicit cross-modal grounding is not sufficient to support robust language-aligned descriptor generation.

Necessity of Stage 1 cross-modal alignment. Removing Stage 1 alignment leads to a clear performance drop across all tasks, with the largest degradation observed on the material descriptor task. In particular, descriptor recall decreases substantially from 54.81% to 17.46%, confirming that explicit alignment between perception and language is especially important for descriptor generation. This result shows that Stage 1 is critical for mapping vision and touch into a shared language-anchored semantic space.

Benefits of full VitaTouch. By combining global visual context, local tactile evidence, and explicit cross-modal alignment, the full VitaTouch model consistently outperforms all ablated variants. The gains are particularly pronounced on the material descriptor task, indicating that multimodal fusion is most beneficial when the model must generate semantically meaningful property descriptions rather than only predict discrete labels.

5.3.4. LoRA-Based Defect Adaptation for Data-Limited Manufacturing Inspection

To further evaluate VitaTouch in data-limited manufacturing inspection, we perform LoRA-based defect adaptation on a separately collected defect-recognition dataset acquired under the same robotic inspection setup. This dataset is used only for downstream Stage 3 adaptation and evaluation. We adopt progressively finer-grained defect label spaces with 2, 3, and 5 categories, and report few-shot adaptation results with the number of labeled training samples per category set to $K \in \{5, 15, 50\}$, while keeping the evaluation set fixed across different K settings. The category design follows a coarse-to-fine defect recognition protocol. The 2-category setting corresponds to coarse defect presence recognition, distinguishing *Defect* from *Normal*. The 3-category setting further introduces defect-type discrimination with three classes: *Defect: Missing*, *Defect: Irregular*, and *Normal*.

The 5-category setting additionally incorporates defect-location information, resulting in a finer label space consisting of *Defect: Missing @Edge*, *Defect: Missing @Local*, *Defect: Irregular @Local*, *Defect: Irregular @Edge*, and *Normal*. Accordingly, the progression from 2 to 3 to 5 categories defines a coarse-to-fine evaluation hierarchy, moving from defect-presence recognition to defect-type recognition and finally to joint defect-type/location recognition.

During Stage 3 adaptation, the perception encoders and Q-Formers are frozen, and only the LoRA parameters inserted into the LLM decoder are updated, enabling parameter-efficient transfer to downstream defect recognition.

Table 3 reports the results of three adaptation settings: vision-only, tactile-only, and the full VitaTouch model with fused vision-tactile inputs. Overall, VitaTouch delivers the best performance across all evaluated settings, and its advantage remains evident under progressively finer-grained defect label spaces. Under the 2-category setting, VitaTouch improves from 90.0% at $K = 5$ to 96.7% at $K = 15$, and further reaches 100.0% at $K = 50$. Under the 3-category setting, the corresponding accuracies are 85.0%, 88.9%, and 96.0%. Under the 5-category setting, VitaTouch attains 72.0%, 85.3%, and 92.0%, respectively.

Compared with the unimodal baselines, the multimodal VitaTouch model shows a clear and consistent advantage, which remains evident as the label space becomes more fine-grained. In particular, the 5-category setting is more challenging because the model must identify both defect type and defect location. Even in this setting, VitaTouch achieves 85.3% accuracy at $K = 15$, outperforming the vision-only and tactile-only variants, which achieve 73.3% and 70.7%, respectively. When $K = 50$, VitaTouch further improves to 92.0%, compared with 86.0% for vision-only and 82.0% for tactile-only adaptation. These results demonstrate that LoRA-based adaptation can effectively leverage limited labeled data while preserving the benefit of multimodal grounding, and that fusing visual and tactile cues is beneficial for robust defect recognition in more fine-grained manufacturing scenarios.

Table 3: LoRA-based defect adaptation results under different numbers of defect categories and labeled training samples per category. Three settings are compared: vision-only, tactile-only, and the full VitaTouch model with fused vision–tactile inputs.

Defect Categories	Train Samples / Category	Vision-only LoRA (Qwen-2.5-VL)	Tactile-only LoRA (Octopi)	VitaTouch (V+T)
2	5	70.0%	60.0%	90.0%
2	15	76.7%	66.7%	96.7%
2	50	82.0%	77.0%	100.0%
3	5	66.7%	66.7%	85.0%
3	15	75.6%	73.3%	88.9%
3	50	86.7%	81.3%	96.0%
5	5	64.0%	56.0%	72.0%
5	15	73.3%	70.7%	85.3%
5	50	86.0%	82.0%	92.0%

5.3.5. Qualitative Results: Inspection-style Outputs

Figure 6 presents representative qualitative outputs of VitaTouch on four inspection-oriented tasks, including material descriptor prediction, hardness classification, roughness classification, and defect decision with brief descriptions. The purpose of these examples is to illustrate the types of responses that VitaTouch can generate under different task settings. As shown in the figure, the model is able to follow the required QA format and produce inspection-style outputs that are aligned with the corresponding prompts. These qualitative examples complement the quantitative results by providing an intuitive view of how VitaTouch expresses property-related predictions and defect-related decisions in operator-facing scenarios.

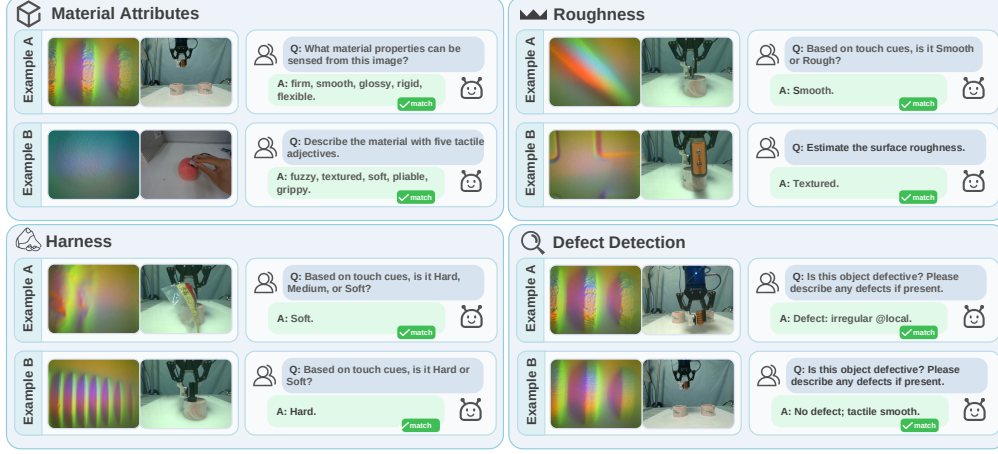


Figure 6: **Qualitative inspection-style outputs of VitaTouch.** Representative examples are shown for material descriptor prediction, hardness classification, roughness classification, and defect decision with brief descriptions.

5.3.6. Proof-of-Concept: Robotic Sorting Demonstration

To illustrate deployment feasibility, we provide a proof-of-concept closed-loop perception-action demonstration for robotic sorting that emulates an inspection-oriented workflow (Fig. 7). A robotic arm grasps each object, captures aligned vision and tactile observations during contact, and then invokes the Stage 3 LoRA-adapted model for defect recognition. Although the model distinguishes multiple defect categories, all defect predictions are mapped to a unified defect action for robotic execution, while normal predictions are mapped to a normal action. The resulting binary decision is used to trigger closed-loop sorting, with defective objects placed into the left bin and normal objects placed into the right bin. This example is intended to illustrate how VitaTouch can be integrated into a robotic inspection loop, rather than to provide a comprehensive evaluation of robotic manipulation performance.

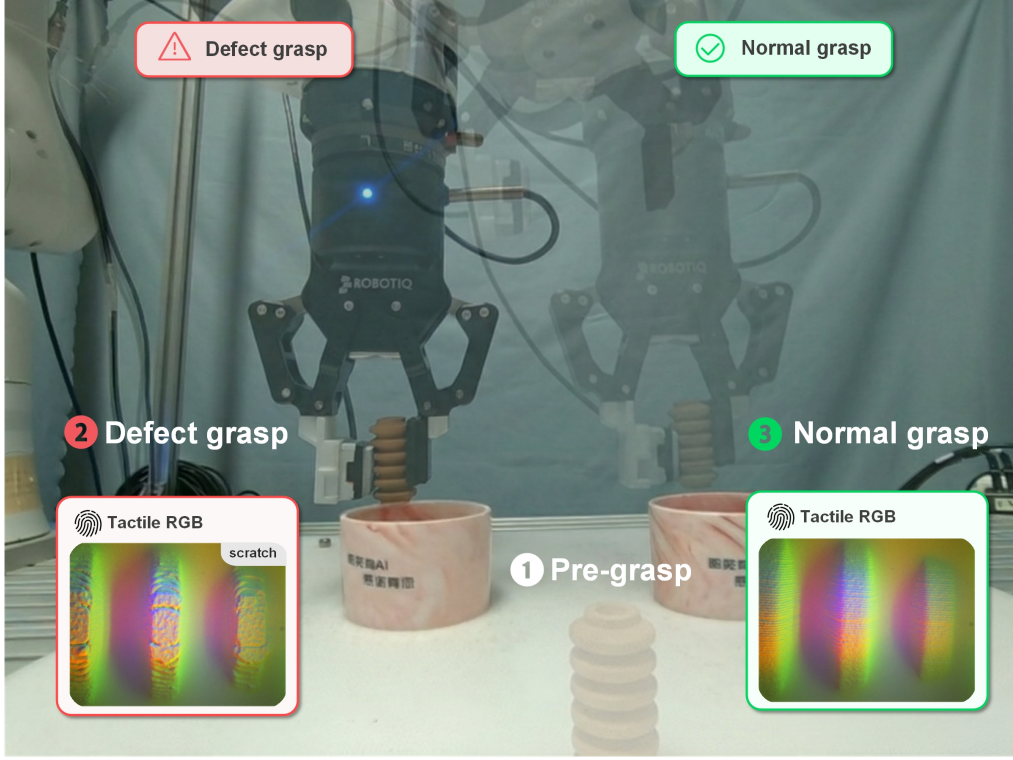


Figure 7: Proof-of-concept robotic sorting demonstration. The robot performs pre-grasp, acquires tactile RGB observations during grasp, predicts defect status using the Stage 3 model, and sorts the object into the defect (left) or normal (right) bin accordingly.

6. Conclusion

We presented **VitaTouch**, a property-aware vision–tactile–language model for manufacturing quality inspection. VitaTouch distills visual and tactile evidence through dual Q-Formers and conditions a frozen large language model, enabling natural-language-driven property understanding and defect-oriented reasoning beyond appearance-only inspection pipelines.

Experiments demonstrate strong effectiveness and transferability. On the public TVL benchmark, VitaTouch achieves the best scores on HCT and TVL among the compared methods. On VitaSet, VitaTouch reaches up to 90.21% hardness accuracy, 77.78% roughness accuracy, and 55.50% descriptor recall on the material descriptor task. In addition, the supplementary semantic-similarity evaluation reaches 0.9009, indicating that the generated descriptors remain semantically well aligned with the human-annotated ref-

erences even when exact lexical overlap is incomplete. Under LoRA-based defect adaptation, VitaTouch maintains strong performance across progressively finer-grained defect label spaces, achieving up to 100.0%, 96.0%, and 92.0% accuracy in 2-, 3-, and 5-category settings, respectively, when 50 labeled samples per category are provided.

These results suggest that property-oriented multimodal grounding is a promising basis for data-limited manufacturing inspection. At the same time, our current evaluation remains limited to 186 objects, curated laboratory conditions, and a proof-of-concept robotic setup. Real deployment in shop-floor environments will require broader validation under environmental variability, more diverse defect distributions, stronger comparisons against prevailing industrial baselines, and more extensive physical experiments. Future work will therefore focus on larger-scale industrial data collection, cross-domain generalization analysis, stronger real-world validation, and deeper integration with practical manufacturing inspection workflows.

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Appendix A. Data Collection Prompts

This appendix presents the prompt templates used during VitaSet annotation with GPT-4o API. All prompts follow a structured format with task-specific rules derived from tactile perception principles.

Appendix A.1. Hardness Classification Prompt

You are a tactile perception expert. You receive a vision image, a tactile image, plus the object name and description.

Classify hardness into ONE label (binary):

- Hard → rigid, barely deforms
- Soft → clearly compressible

Key cues (VERY IMPORTANT):

- Primary rule: If the tactile contact boundary/edges look sharp, crisp, and well-defined, the object is likely Hard.
- Primary rule: If the tactile contact boundary/edges look blurred, diffuse, or not clearly defined, the object is likely Soft.
- Use the vision image as a consistency check:
 - * If tactile edges are sharp AND the vision image shows little/no visible deformation, choose Hard (high confidence).
 - * If tactile edges are blurred AND the vision image shows obvious deformation/indentation, choose Soft (high confidence).
 - * If cues conflict (e.g., sharp edges but visible deformation), still follow the tactile-edge rule as the primary signal, but use the object name/description to break ties.

Use visual cues, gel deformation, and text context.

Object: {object_name}

Description: {description}

Reply with exactly one word: Hard or Soft.

Appendix A.2. Roughness Classification Prompt

You analyze surface texture from a vision image, a tactile image, and the object info below.

Choose ONE label:

- Smooth → even, flat surface with little/no bumps
- Textured → regular/repeating raised pattern (structured bumps/ridges)
- Rough → clearly bumpy/irregular micro-relief, but WITHOUT a regular pattern

Key cues (VERY IMPORTANT):

- If the contact surface looks flat and even, choose Smooth.
- If there are small bumps/relief that are obvious but irregular and non-repeating, choose Rough.
- If the raised pattern is regular/repeating (organized texture), choose Textured.

Consider tactile detail, surface micro-structure, and the description.

Object: {object_name}

Description: {description}

Reply with exactly one word: Smooth, Textured, or Rough.

Appendix A.3. Material Property Description Prompt

You are a tactile perception expert. You receive: (1) a vision image, (2) a tactile image, (3) the object name and a short description.

Task: Select ~5 tactile adjectives (prefer EXACTLY 5) that best describe the surface/material properties.

IMPORTANT: You MUST choose ONLY from this fixed vocabulary (case-insensitive), and you MUST NOT invent new words:
hard, firm, soft, cushioned, rigid, flexible, smooth, rough, grainy, fibrous, uneven, patterned, glossy, matte, slippery, sticky, grippy, metallic, plastic, woven

Rules:

- Output 3 to 5 words (prefer 5).
- Use only the words from the list above.
- Do not output temperature, color, size, shape, or object name.
- Comma + space separated. Lowercase. End with a period.
- Choose the closest matches based on BOTH images and the text info.

Object: {object_name}

Description: {description}

Output:

Appendix A.4. Human Verification Protocol

All GPT-4o generated labels underwent complete object-level human verification across all 186 objects. Reviewers corrected inconsistencies, implausible descriptions, and format violations to ensure label quality and inter-annotator agreement.

References

- [1] V. Azamfirei, F. Psarommatis, Y. Lagrosen, Application of automation for in-line quality inspection, a zero-defect manufacturing approach, *Journal of Manufacturing Systems* 67 (2023) 1–22.
- [2] L. Lu, J. Hou, S. Yuan, X. Yao, Y. Li, J. Zhu, Deep learning-assisted real-time defect detection and closed-loop adjustment for additive manufacturing of continuous fiber-reinforced polymer composites, *Robotics and Computer-Integrated Manufacturing* 79 (2023) 102431.
- [3] A. M. Reis, A. Dall-Orsoletta, E. Nunes, L. Costa, S. Sousa, Quality costs and industry 4.0: inspection strategy modelling and reviewing, *The International Journal of Advanced Manufacturing Technology* 136 (2025) 3883–3897.
- [4] J. Liu, G. Xie, J. Wang, S. Li, C. Wang, F. Zheng, Y. Jin, Deep industrial image anomaly detection: A survey, *Machine Intelligence Research* 21 (2024) 104–135.
- [5] P. Bergmann, M. Fauser, D. Sattlegger, C. Steger, MVTec AD – a comprehensive real-world dataset for unsupervised anomaly detection, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 9592–9600.
- [6] Y. Ma, J. Yin, F. Huang, Q. Li, Surface defect inspection of industrial products with object detection deep networks: a systematic review, *Artificial Intelligence Review* 57 (2024) 333.
- [7] T. Li, Y. Yan, C. Yu, J. An, Y. Wang, G. Chen, A comprehensive review of robot intelligent grasping based on tactile perception, *Robotics and Computer-Integrated Manufacturing* 90 (2024) 102792.
- [8] Y.-H. Xin, K.-M. Hu, R.-J. Xiang, Y.-L. Gao, J.-F. Zhou, G. Meng, W.-M. Zhang, Vision-based tactile sensing: From performance parameters to device design, *Applied Physics Reviews* 12 (2025) 021312.
- [9] R. Feng, J. Hu, W. Xia, T. Gao, A. Shen, Y. Sun, B. Fang, D. Hu, AnyTouch: Learning unified static-dynamic representation across multiple visuo-tactile sensors, in: *International Conference on Learning Representations (ICLR)*, 2025.

- [10] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning transferable visual models from natural language supervision, in: Proceedings of the 38th International Conference on Machine Learning, 2021, pp. 8748–8763.
- [11] J. Li, D. Li, S. Savarese, S. Hoi, BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models, in: Proceedings of the 40th International Conference on Machine Learning, 2023, pp. 19730–19742.
- [12] W. Dai, J. Li, D. Li, A. M. H. Tiong, J. Zhao, W. Wang, B. Li, P. Fung, S. Hoi, InstructBLIP: Towards general-purpose vision-language models with instruction tuning, in: Advances in Neural Information Processing Systems, volume 36, 2023, pp. 49250–49267.
- [13] H. Liu, C. Li, Q. Wu, Y. J. Lee, Visual instruction tuning, in: Advances in Neural Information Processing Systems, volume 36, 2023, pp. 34892–34916.
- [14] L. Fu, G. Datta, H. Huang, W. C.-H. Panitch, J. Drake, J. Ortiz, M. Mukadam, M. Lambeta, R. Calandra, K. Goldberg, A touch, vision, and language dataset for multimodal alignment, in: Proceedings of the 41st International Conference on Machine Learning, 2024, pp. 14080–14101.
- [15] S. Yu, K. Lin, A. Xiao, J. Duan, H. Soh, Octopi: Object property reasoning with large tactile-language models, in: Robotics: Science and Systems, 2024.
- [16] X. Jiang, J. Li, H. Deng, Y. Liu, B.-B. Gao, Y. Zhou, J. Li, C. Wang, F. Zheng, MMAD: A comprehensive benchmark for multimodal large language models in industrial anomaly detection, in: International Conference on Learning Representations (ICLR), 2025.
- [17] T. Defard, A. Setkov, A. Loesch, R. Audigier, PaDiM: a patch distribution modeling framework for anomaly detection and localization, in: Computer Analysis of Images and Patterns, 2021, pp. 475–489.
- [18] V. Zavrtanik, M. Kristan, D. Skočaj, DRAEM – A discriminatively trained reconstruction embedding for surface anomaly detection, in:

Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2021, pp. 8330–8339.

- [19] K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, P. Gehler, Towards total recall in industrial anomaly detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 14318–14328.
- [20] Y. Zou, J. Jeong, L. Pemula, D. Zhang, O. Dabeer, SPot-the-difference self-supervised pre-training for anomaly detection and segmentation, in: Computer Vision – ECCV 2022, 2022, pp. 392–408.
- [21] C. Wang, W. Zhu, B.-B. Gao, Z. Gan, J. Zhang, Z. Gu, S. Qian, M. Chen, L. Ma, Real-IAD: A real-world multi-view dataset for benchmarking versatile industrial anomaly detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, pp. 22883–22892.
- [22] J. Jeong, Y. Zou, T. Kim, D. Zhang, A. Ravichandran, O. Dabeer, Win-CLIP: Zero-/few-shot anomaly classification and segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023, pp. 19606–19616.
- [23] X. Li, Z. Zhang, X. Tan, C. Chen, Y. Qu, Y. Xie, L. Ma, PromptAD: Learning prompts with only normal samples for few-shot anomaly detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, pp. 16838–16848.
- [24] Q. Zhou, G. Pang, Y. Tian, S. He, J. Chen, AnomalyCLIP: Object-agnostic prompt learning for zero-shot anomaly detection, in: International Conference on Learning Representations (ICLR), 2024.
- [25] Z. Qu, X. Tao, M. Prasad, F. Shen, Z. Zhang, X. Gong, G. Ding, VCP-CLIP: A visual context prompting model for zero-shot anomaly segmentation, in: Computer Vision – ECCV 2024, 2024, pp. 301–317.
- [26] Y. Cao, J. Zhang, L. Frittoli, Y. Cheng, W. Shen, G. Boracchi, AdaCLIP: Adapting clip with hybrid learnable prompts for zero-shot anomaly detection, in: Computer Vision – ECCV 2024, 2024, pp. 55–72.

- [27] S. Li, J. Cao, P. Ye, Y. Ding, C. Tu, T. Chen, ClipSAM: Clip and sam collaboration for zero-shot anomaly segmentation, *Neurocomputing* 618 (2025) 129122.
- [28] W. Ma, X. Zhang, Q. Yao, F. Tang, C. Wu, Y. Li, R. Yan, Z. Jiang, S. K. Zhou, AA-CLIP: Enhancing zero-shot anomaly detection via anomaly-aware clip, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 4744–4754.
- [29] T. Gong, Q. Chu, B. Liu, W. Zhou, N. Yu, FE-CLIP: Frequency enhanced clip model for zero-shot anomaly detection and segmentation, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 21220–21230.
- [30] Z. Gu, B. Zhu, G. Zhu, Y. Chen, M. Tang, J. Wang, AnomalyGPT: Detecting industrial anomalies using large vision-language models, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, 2024, pp. 1932–1940.
- [31] W. Yuan, S. Dong, E. H. Adelson, GelSight: High-resolution robot tactile sensors for estimating geometry and force, *Sensors* 17 (2017) 2762.
- [32] M. Lambeta, P.-W. Chou, S. Tian, B. Yang, B. Maloon, V. R. Most, D. Stroud, R. Santos, A. Byagowi, D. Jayaraman, R. Calandra, DIGIT: A novel design for a low-cost compact high-resolution tactile sensor with application to in-hand manipulation, *IEEE Robotics and Automation Letters* 5 (2020) 3838–3845.
- [33] A. Agarwal, A. Ajith, C. Wen, V. Stryzheus, B. Miller, M. Chen, M. K. Johnson, J. L. Susa Rincon, J. Rosca, W. Yuan, Robotic defect inspection with visual and tactile perception for large-scale components, in: *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 10110–10116.
- [34] C. Finn, P. Abbeel, S. Levine, Model-agnostic meta-learning for fast adaptation of deep networks, in: *Proceedings of the 34th International Conference on Machine Learning*, volume 70, 2017, pp. 1126–1135.

- [35] J. Snell, K. Swersky, R. Zemel, Prototypical networks for few-shot learning, in: Advances in Neural Information Processing Systems, volume 30, 2017.
- [36] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, LoRA: Low-rank adaptation of large language models, in: International Conference on Learning Representations (ICLR), 2022.